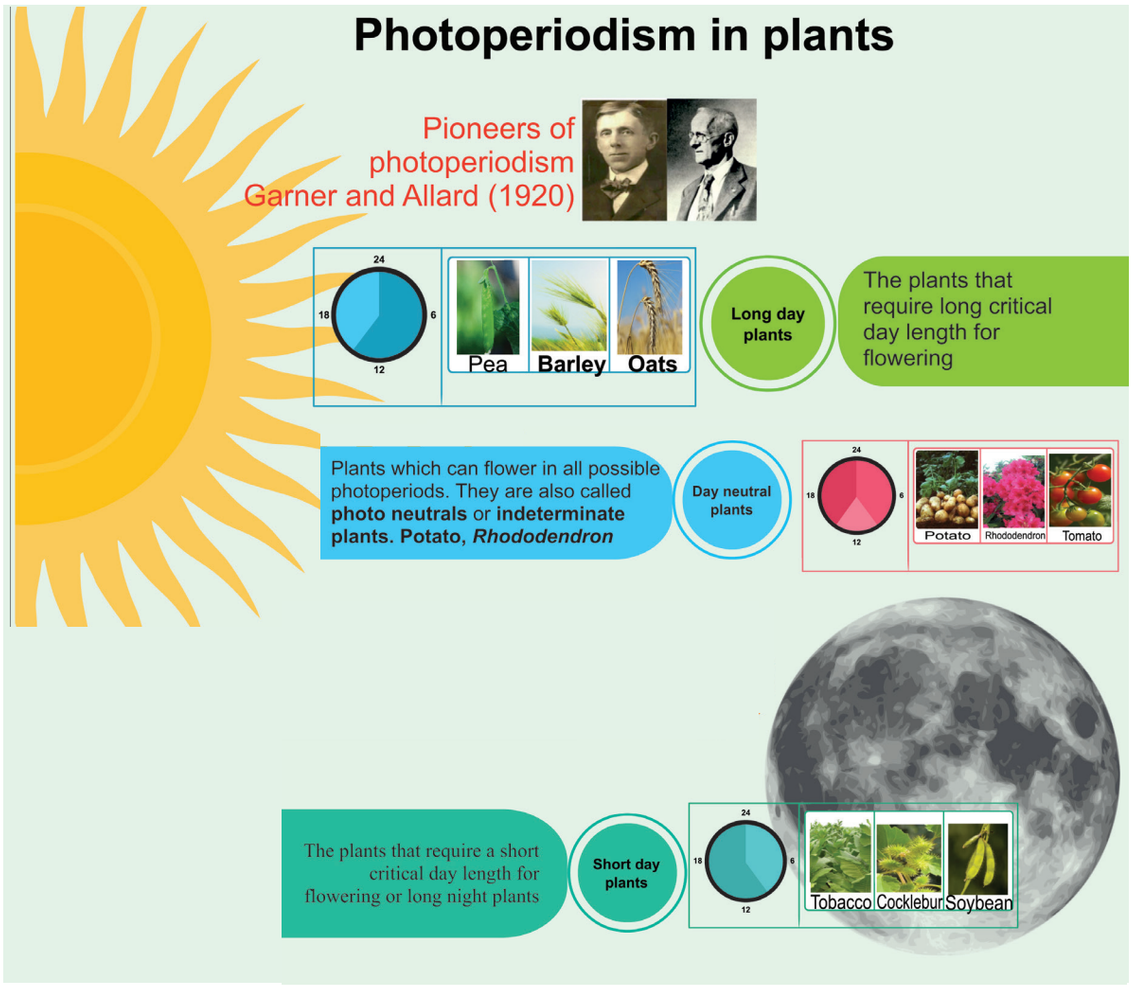


(1959) named this pigment and it exists in two interconvertible forms: (i) red light absorbing pigment which is designated as P_r and (ii) far red light absorbing pigment which is designated as P_{fr}. The P_r form absorbs red light in 660nm and changes to $P_{fr.}$ The P_{fr} form absorbs far red light in 730nm and changes to P_r. The P_r form is biologically inactive and it is stable whereas P_{fr} form is biologically active and it is very unstable. In short day plants, P_r promotes flowering and P_{fr} inhibits the flowering whereas in long day plants flowering is promoted by P_{fr} and inhibited by P_r form. P_{fr} is always associated with hydrophobic area of membrane systems while P_r is found in diffused state in the cytoplasm. The interconversion of the two forms of phytochrome is mainly involved in flower induction and also additionally plays a role in seed germination and changes in membrane conformation.

15.4 Vernalization (*Vernal* – Spring Like)

Besides photoperiod certain plants require a low temperature exposure in their earlier stages for flowering. Many species of biennials and perennials are induced to flower by low temperature exposure (0°C to 5°C). This process is called **Vernalization.** The term Vernalization was first used by **T. D. Lysenko** (1938).

1. Mechanism of Vernalization:

Two main theories to explain the mechanism of vernalization are:

i. Hypothesis of phasic development
ii. Hypothesis of hormonal involvement

i. Hypothesis of phasic development

XIth_BIO-BOTANY_Ch_15-EM.indd 286 5/17/2020 11:30:33 PM